



Retrieval-Augmented Garment Generation with Diffusion Models

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Course: Neural Networks

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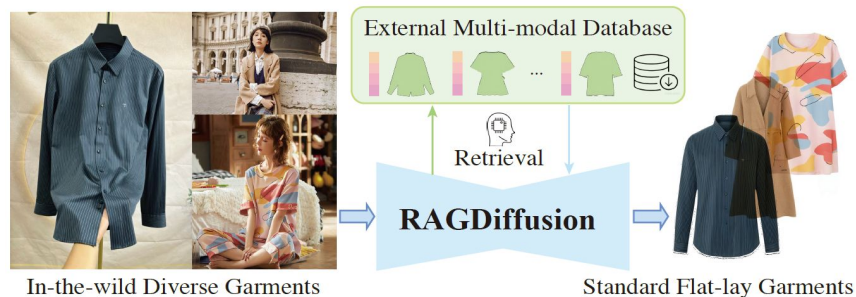


Motivation & Objectives

The problem: Standard diffusion models fail to synthesize standard flat-lay garments from in-the-wild diverse images.

- “Structural Hallucinations”: artifacts such as incorrect patterns or shapes, distorted necklines produced by the model in the generation.

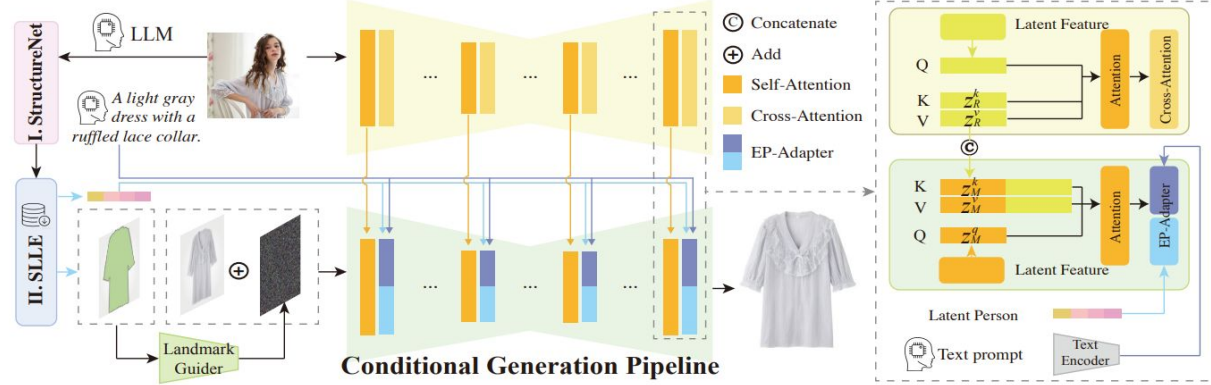
The solution: Retrieval Augmented Generation (RAG), augment the model by retrieving a reliable structural guidance from an external database to reduce hallucinations and improve structural accuracy.



Project Goal: investigate the RAG mechanism by replicating the framework proposed in *RAGDiffusion* [1] and adapting it to settings with limited computational resources



Methodology: The RAG Pipeline



The garment generation is conditioned on 4 inputs:

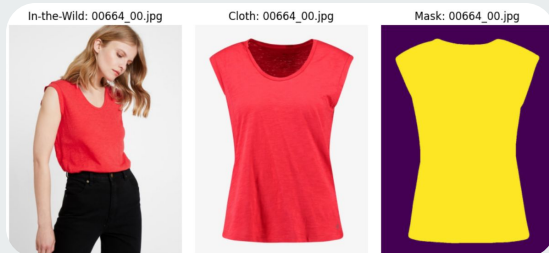
- 1. Structure Embeddings:**
 - a. Computed by StructureNet from the fusion of visual CLIP features + LLM attributes
 - b. Refined via SLLE
- 2. Garment silhouette:** retrieved from the Retrieval Database, acting as an hard constraint on the final garment shape.
- 3. Prompt:** generated by a LLM, provides explicit semantic context such as color, pattern, and fabric type.
- 4. Reference Latents:** the clean latent of the original in-the-wild image produced with the SDXL VAE, serving as the source of high-frequency details.



Implementation

Dataset VITON-HD:

~13k high resolution images (1024x768). Paired garment and in-the-wild images with associated garment masks



Retrieval database D

- Consists of pairs $\rightarrow$ (Standard embedding, Mask)
- Clustering: DBSCAN ($\epsilon=0.4$) to group structurally similar items, remove redundancy and outliers.
- Downsampling: Cluster-Aware Sampling (proportionally to the cluster size) to obtain a database of 2000 pairs.

This maintains the structural diversity while reducing retrieval latency and memory usage.

StructureNet Training:

- Designed to map garment image from 2 distinct domain (In-the-Wild - Flat-Lay) into a shared embedding space.
- Trained using Contrastive Loss (InfoNCE) on a representative subset of 100 pairs.

Model Configuration	Batch Size	Epochs	CLIP Status	Recall@1
StructureNet (Baseline)	4	50	Frozen	0.65
StructureNet (Baseline)	4	25	Frozen	0.75
StructureNet v3	4	50	Fine-Tuned	0.70
StructureNet v2 (Selected)	8	50	Fine-Tuned	0.85



Testing the performance on a retrieval task (given the in-the wild image return the k best match in the test set)

Implementation

Hard constraint (Silhouette mask)

1. **Silhouette retrieval:** replaced SLLE which produced blurry masks, with a Cohesion-based selection (choose the single candidate with highest average IoU overlap from the k retrieved neighbors)
2. **Injection:** I integrated the pretrained ControlNet-Canny (SDXL 1.0) using the binary silhouette as strong edge map which the Canny model is expert at interpreting.



Metric	Result (n=20)	Interpretation
Mean IoU	0.8244	High overall geometric overlap
Precision (IoU > 0.85)	55.00%	Successful high-fidelity retrieval for 11/20 cases

Soft constraint (Structural embedding)

- The EP-Adapter first projects the embeddings into the dimensional space of the SDXL text encoder and then adds a parallel Cross-Attention layer in the U-Net for joint text-structure conditioning.
- Adapter weights remain untrained due to computational constraints.
- Generation quality is currently mainly driven by the geometric constraint (ControlNet) and semantic control (LLM).

Analysis and Visual results

Due to significant computational constraints, large-scale quantitative benchmarking over thousands of samples was not feasible. Instead, this project focused on **component-level validation** and **perceptual faithfulness**, providing insight on the effectiveness of the proposed multi-level conditioning.

- **Perceptual Faithfulness:** computed Learned Perceptual Image Patch Similarity (LPIPS), which measures the perceptual distance between generated output and ground truth, across a test set of 12 samples, achieving a Mean LPIPS of 0.706 (Lower is better).



Observations:

- Large logos and macro structures are quite preserved while fine textures and small, repetitive details aren't well reproduced.
- The pipeline functions correctly end-to-end, but the overall fidelity remains limited by computational constraints. .



Conclusion

The primary objective of this project was not to achieve state of the art performance, but to understand the mechanisms of Retrieval-Augmented Generation (RAG) within the context of diffusion-based image synthesis.

I successfully validated the multi-level conditioning pipeline, understanding how external knowledge (Retrieval) can effectively mitigate structural hallucinations.

Future Work

- **Train EP-Adapter:** Unfreeze weights to enable texture transfer from StructureNet.
- **Implement Decoding Faithfulness:** Replace the standard SDXL VAE (which suffers from high compression artifacts) with the 16-channel FLUX VAE and align the 16-channel VAE latent space with the 4-channel SDXL U-Net.
- **Expand Dataset Diversity:** to improve generalization beyond human-worn inputs.





References

- **Papers:**

- [1] Yuhao Li, Xianfeng Tan, Wenxiang Shang, Yubo Wu, Jian Wang, Xuanhong Chen, Yi Zhang, Ran Lin, Bingbing Ni. (2025). **RAG-Diffusion: Faithful Cloth Generation via External Knowledge Assimilation**. arXiv preprint arXiv:2411.19528v2. (Primary architectural framework).
- [2] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. (2023). **SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis**. arXiv preprint arXiv:2307.01952. (Backbone generative model).
- [3] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. (2023). **Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond**. arXiv preprint arXiv:2308.12966. (Used for semantic attribute extraction and prompt generation).
- [4] Hu Ye, Jun Zhang, Sibor Liu, Xiao Han, Wei Yang. (2023). **IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models**. arXiv preprint arXiv:2308.06721. (Source for the Perceiver Resampler and the architectural basis of the EP-Adapter).
- [5] Li Hu, Xin Gao, Peng Zhang, Ke Sun, Bang Zhang, and Liefeng Bo. (2023). **Animate anyone: Consistent and controllable image-to-video synthesis for character animation**. arXiv preprint arXiv:2311.17117. (Methodological source for ReferenceNet and spatial feature injection).
- [6] Seunghwan Choi, Sunghyun Park, Minsoo Lee, Jaegul Choo. (2021). **VITON-HD: High-Resolution Virtual Try-On via Misalignment-Aware Normalization**. arXiv preprint arXiv:2103.16874. (Dataset source via Kaggle).

- **Slides:**

- <https://github.com/pietro-nardelli/sapienza-ppt-template>
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Thank you for the attention!

